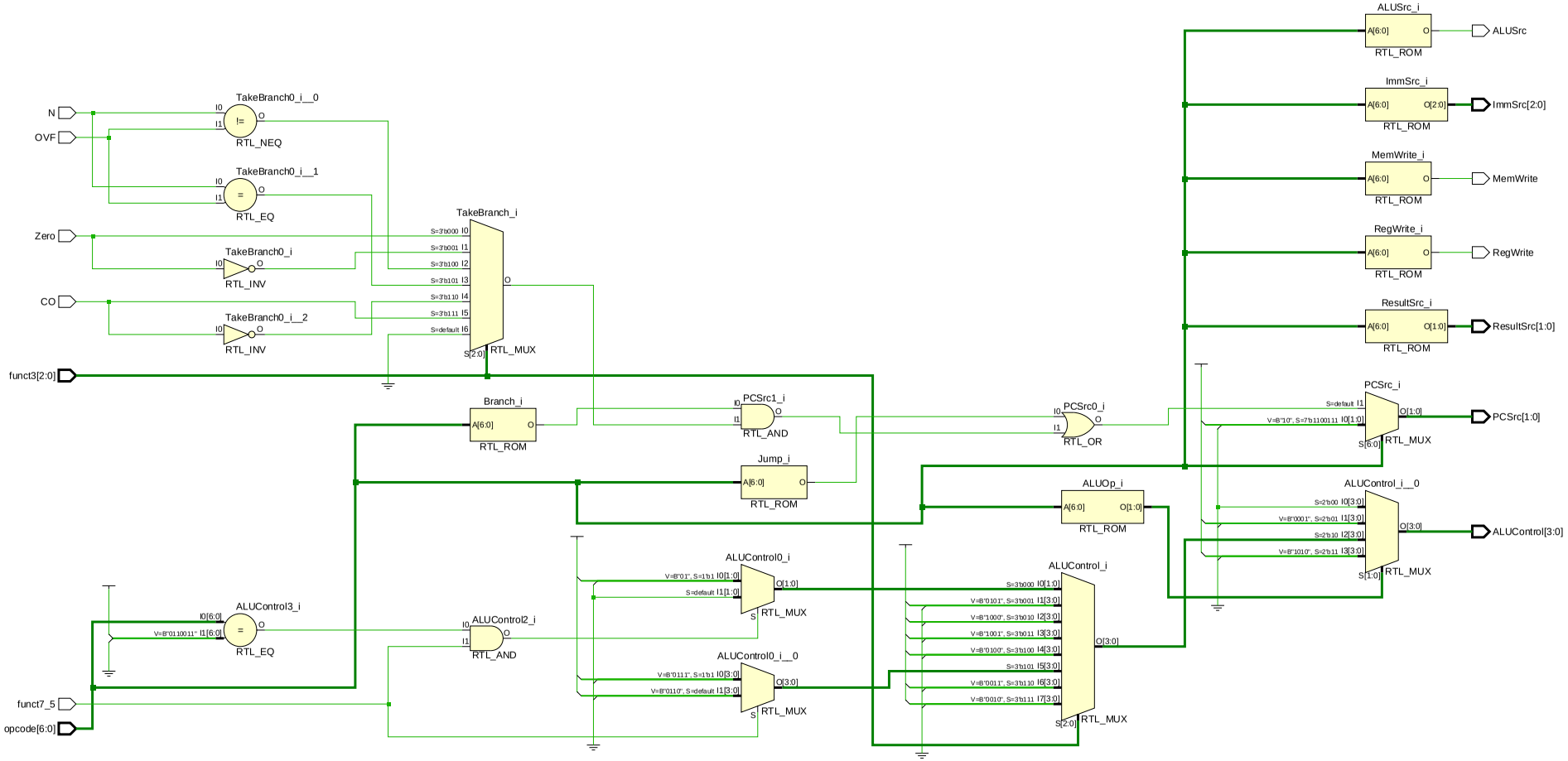
N
OVF
Zero
CO
funct3[2:0]
funct7_5
opcode[6:0]
TakeBranch0_i__0
!=
RTL_NEQ
TakeBranch0_i__1
=
RTL_EQ
TakeBranch0_i
RTL_INV
TakeBranch0_i__2
RTL_INV
TakeBranch_i
S=3'b000 I0
S=3'b001 I1
S=3'b100 I2
S=3'b101 I3
S=3'b110 I4
S=3'b111 I5
S=default I6
S[2:0]
RTL_MUX
ALUControl3_i
I0[6:0]
V=B"0110011" I1[6:0]
=
RTL_EQ
ALUControl2_i
RTL_AND
Branch_i
A[6:0]
RTL_ROM
PCSrc1_i
RTL_AND
Jump_i
A[6:0]
RTL_ROM
ALUControl0_i
V=B"01", S=1'b1 I0[1:0]
S=default I1[1:0]
O[1:0]
RTL_MUX
ALUControl0_i__0
V=B"111", S=1'b1 I0[3:0]
V=B"0110", S=default I1[3:0]
O[3:0]
RTL_MUX
PCSrc0_i
RTL_OR
ALUOp_i
A[6:0]
O[1:0]
RTL_ROM
ALUControl_i
S=3'b000 I0[1:0]
V=B"0101", S=3'b001 I1[3:0]
V=B"1000", S=3'b010 I2[3:0]
V=B"1001", S=3'b011 I3[3:0]
V=B"0100", S=3'b100 I4[3:0]
S=3'b101 I5[3:0]
V=B"0011", S=3'b110 I6[3:0]
V=B"0010", S=3'b111 I7[3:0]
O[3:0]
S[2:0]
RTL_MUX
ALUSrc_i
A[6:0]
O
RTL_ROM
ALUSrc
ImmSrc_i
A[6:0]
O[2:0]
RTL_ROM
ImmSrc[2:0]
MemWrite_i
A[6:0]
O
RTL_ROM
MemWrite
RegWrite_i
A[6:0]
O
RTL_ROM
RegWrite
ResultSrc_i
A[6:0]
O[1:0]
RTL_ROM
ResultSrc[1:0]
PCSrc_i
S=default I1
V=B"10", S=7'b1100111 I0[1:0]
O[1:0]
S[6:0]
RTL_MUX
PCSrc[1:0]
ALUControl_i__0
S=2'b00 I0[3:0]
V=B"0001", S=2'b01 I1[3:0]
S=2'b10 I2[3:0]
V=B"1010", S=2'b11 I3[3:0]
O[3:0]
S[1:0]
RTL_MUX
ALUControl[3:0]